Anderson localisation with self-avoiding-walks

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1 Introduction

The Anderson localization model was introduced by the physicist P.W. Anderson in 1958 to explain the quantum mechanical effects of disorder in materials. Anderson discusses the behaviour of electrons in dirty crystals, which is the quantum mechanical analougue of a random walk in a random environment. The model considers a d-dimensional grid, in which each site represents an atom. From an heuristical point of view, we also consider electrons that jump from atom to atom and are subject to an external random potential, which influences their motion in the material. When these particles cannot jump , it indicates the absence of quantum transport.This phenomenon is called Anderson localisation. This model deals with conditions for which the electrons stay trapped in a specific region and don't generate electric conduction, defining an insulator. This is in sharp contrast to the behaviour in ideal crystals, which are always conductors.
Anderson's work, and that of N.F.Mott and J.H. van Vleck, won the 1977 physics Nobel prize for "their fundamental theoretical investigations of the electronic structure of magnetic and disordered systems". Since then, the study of random operators has become an important field of mathematical physics, which has led to a tremendous amount of research activity and many mathematical results.
The two important methods to prove Anderson localization are the *multiscale analysis method* (MSA), developed by Froehlich and Spencer [] in 1983, and the *fractional moment method* (FMM) by Aizenman and Molchanov in 1993. We will deal with the second one, which is more elementary and gives stronger results on *dynamical localization* [Chapter 2.2].

structure of the thesis

Prerequisites

It is useful to have some familiarity with measure theory, basic probabilistic concepts such as indipendence and continuous distributions, and foundations of the theory of linear operators in Hilbert spaces, up to the spectral theorem for self adjoint operators and consequences (as spectral types).

2 The Anderson Model

Transport phenomena in a disordered material are often described via discrete random Schrödinger operator. In quantum mechanics, it corresponds to the classical total energy decomposing into kinetic and potential energy. On the lattice $\mathbb{Z}^d$, it takes the form of an infinite random matrix called the *Hamiltonian*:

$$H_\omega = -\Delta + \lambda V_\omega.$$

where $-\Delta$ is a deterministic matrix and λV_ω is a diagonal matrix where V_ω has random entries and λ is the disorder parameter.
This operator acts on the Hilbert space

$$\ell^2(\mathbb{Z}^d) = \{u : \mathbb{Z}^d \to \mathbb{C} : \sum_{n \in \mathbb{Z}^d} |u(n)|^2 < \infty\}$$

with inner product $\langle u, v\rangle = \sum_n \overline{u(n)} v(n)$.

In quantum mechanics, a physical state describing a single particle (such as an electron) is represented by a normalized wave function $\psi \in \ell^2(\mathbb{Z}^d)$ satisfying

$$\|\psi\|_{\ell^2}^2 = \sum_{x \in \mathbb{Z}^d} |\psi(x)|^2 = 1. \tag{2.1}$$

The quantity $P(x) = |\psi(x)|^2$ defines a probability mass function on the lattice $\mathbb{Z}^d$: it represents the probability of finding the particle localized at site $x \in \mathbb{Z}^d$.
This functions are the solutions to the time-evolving Schrödinger operator $H_\omega \phi(t) = \partial_t \phi(t)$. Because the time evolution operator e^{-itH_ω} is unitary on $\ell^2(\mathbb{Z}^d)$, the total probability is strictly conserved at all times $t \in \mathbb{R}$:

$$\|\psi(t)\|_{\ell^2}^2 = \|e^{-itH_\omega}\psi_0\|_{\ell^2}^2 = \|\psi_0\|_{\ell^2}^2 = 1. \tag{2.2}$$

In particular, if the particle is initially placed at a specific site $y \in \mathbb{Z}^d$ with $\psi(0) = \delta_y$, the transition probability to find the particle at site $x \in \mathbb{Z}^d$ at time t is given by the squared transition amplitude:

$$P(y \to x; t) = |\langle \delta_x, e^{-itH_\omega} \delta_y \rangle|^2. \tag{2.3}$$

2.1 The Hamiltonian

The Descrete Laplacian is the descrete analogue of the usual negative Laplacian $(-\sum_j \partial^2/\partial x_j^2)$. For the functions $u \in \ell^2(\mathbb{Z}^d)$ it is defined as:

$$-\Delta u(x) := \sum_{k \in \mathbb{Z}^d, |x-k|=1} u(x) - u(k) \quad \text{for } x \in \mathbb{Z}^d. \tag{2.4}$$

where $| * |$ denotes the ℓ^2 norm.

The discrete Laplacian represents the kinetic term and connects neighbour sites on the lattice, such that the electron can transition from one to another.

For each site x, there are $2d$ possible k such that $|x-k| = 1$. Thus, we can define an infinte matrix such that:

$$(-\Delta)_{x,k} = \begin{cases} 2d & \text{if } x = k \quad \text{(on the diagonal)} \\ -1 & \text{if } |x-k| = 1 \\ 0 & \text{otherwise} \end{cases}$$

For $u \in \ell^2(\mathbb{Z}^d)$, $-\Delta u$ is summable in ℓ^2, bounded, and self-adjoint.

The first claim (summability) is shown by:

$$\begin{aligned}
\sum_{x \in \mathbb{Z}^d} |\Delta u(x)|^2 &= \sum_{x \in \mathbb{Z}^d} \left| \sum_{k \in \mathbb{Z}^d, |x-k|=1} \big(u(x) - u(k)\big) \right|^2 \\
&\leq 2d \sum_{x \in \mathbb{Z}^d} \left(\sum_{|x-k|=1} |u(x) - u(k)|^2 \right) \\
&\leq 2d \sum_{x \in \mathbb{Z}^d} \left(2 \sum_{|x-k|=1} \big(|u(x)|^2 + |u(k)|^2\big) \right) \\
&\leq 2d \cdot 2d \cdot 4 \|u\|_2^2 < \infty
\end{aligned}$$

where we used the bound over the number of neighbors, triangle inequality, Cauchy Schwarz and the invariance under translation of the ℓ^2 norm. The boundedness comes from:

$$\frac{\|\Delta u\|_2^2}{\|u\|_2^2} \leq 16d^2 \implies \|\Delta\|_2 \leq 4d.$$

To prove self-adjointness, we only need to show symmetry (given $-\Delta$ is bounded). For $u, v \in \ell^2(\mathbb{Z}^d)$ and $k \in \mathbb{Z}^d$:

$$\langle u, -\Delta v\rangle = \sum_{x\in\mathbb{Z}^d} \overline{u(x)}(-\Delta v(x)) = \sum_{x\in\mathbb{Z}^d} \overline{u(x)} \left(\sum_{|x-k|=1} v(x) - v(k) \right)$$

$$= \sum_{x\in\mathbb{Z}^d} \left(\overline{u(x)} 2d v(x) - \sum_{|x-k|=1} \overline{u(x)} v(k) \right) = \sum_{x\in\mathbb{Z}^d} 2d\overline{u(x)}v(x) - \sum_{x\in\mathbb{Z}^d} \sum_{|x-k|=1} \overline{u(k)}v(x)$$

$$= \sum_{x\in\mathbb{Z}^d} v(x) \left(\sum_{|x-k|=1} \overline{u(x)} - \overline{u(k)} \right) = \sum_{x\in\mathbb{Z}^d} v(x)\overline{(-\Delta u(x))} = \langle -\Delta u, v\rangle.$$

The spectrum of a bounded, self-adjoint operator is real-valued, ad in this case equal to the closed interva $[0, 4d]$. In this range, we find the levels of kinetic energy that an electron owns. In order to understand this result better, let's consider the discrete Fourier transform:

$$\mathcal{F} \colon \ell^2(\mathbb{Z}^d) \longrightarrow L^2([0, 2\pi)^d)$$

such that:

$$(\mathcal{F}u)(x) := \lim_{N\to\infty} (2\pi)^{-\frac{d}{2}} \sum_{|n|\leq N} u(n)\, e^{-ix\cdot n}, \qquad x \in [0, 2\pi)^d.$$

We consider now calculations for $u \in \ell^1(\mathbb{Z}^d)$. For ℓ^1 is a dense subset of ℓ^2 and $\mathcal{F}$ unitary, we can extend the results by continuity for all $u \in \ell^2(\mathbb{Z}^d)$

$$
\begin{aligned}
(\mathcal{F}(-\Delta u))(x) &= (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} (-\Delta u)(n)\, e^{-ix\cdot n} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} \left(\sum_{|k-n|=1} \big(u(n)-u(k)\big) \right) e^{-ix\cdot n} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} \left(2d\, u(n) - \sum_{|k-n|=1} u(k) \right) e^{-ix\cdot n} \\
&= 2d\,(\mathcal{F}u)(x) - (2\pi)^{-\frac{d}{2}} \sum_{n\in\mathbb{Z}^d} \sum_{|k-n|=1} u(k)\, e^{-ix\cdot n} \\
&= 2d\,(\mathcal{F}u)(x) - (2\pi)^{-\frac{d}{2}} \sum_{k\in\mathbb{Z}^d} u(k) \sum_{|m|=1} e^{-ix\cdot(k+m)} \quad (\text{for } m = n-k) \\
&= 2d\,(\mathcal{F}u)(x) - \left((2\pi)^{-\frac{d}{2}} \sum_{k\in\mathbb{Z}^d} u(k)\, e^{-ix\cdot k} \right) \sum_{|m|=1} e^{-ix\cdot m} \\
&= 2d\,(\mathcal{F}u)(x) - (\mathcal{F}u)(x) \sum_{j=1}^{d} \big(e^{-ix_j} + e^{ix_j}\big) \\
&= \left(2\sum_{j=1}^{d} (1-\cos(x_j)) \right) (\mathcal{F}u)(x).
\end{aligned}
$$

We obtain the multiplication operator:

$$
\mathcal{F}\Delta\mathcal{F}^{-1} = 2\sum_{j=1}^{d}(1-\cos(x_j))
$$

on $L^2([0,2\pi)^d)$ in the variable $x = (x_1, \ldots.., x_d)$. We notice that the Laplacian doesn't have eigenvalues in $\ell^2(\mathbb{Z}^d)$. Another similarity of Δ with the continuum Laplacian is that it has plane waves as generalized eigenfunctions. To see this, let $x \in [0,2\pi)^d$ and set

$$
\phi_x(n) := e^{-in\cdot x}
$$

While $\phi_x \notin \ell^2(\mathbb{Z}^d)$, Δ acts on it as

$$
\begin{aligned}
(\Delta\phi_x)(n) &= \sum_{|k|=1}(e^{-in\cdot x} - e^{-i(n+k)\cdot x}) \\
&= 2d\phi_x(n) - \left(\sum_{j=1}^{d} 2\cos(x_j)\right)\phi_x(n) \quad = \left(2\sum_{j=1}^{d}(1-\cos(x_j))\right)\phi_x.
\end{aligned}
$$

Thus ϕ_x is a bounded generalized normalized eigenfunction of Δ to the spectral value $2\sum_{j=1}^{d}(1-\cos(x_j))$. The spectrum is continuous and bounded in $[0, 2\dot{(}2d)]$.

The Random Potential is expressed as the multiplication operator $V_\omega(n) = \omega_n$. It is a diagonal matrix, where $\omega = (\omega_n)_{n\in\mathbb{Z}^d}$ is the set of the indipendent identically distributed real-valued random entries.

Definition 2.1 (i.i.d. random variables)**.** $(\omega_n)_{n\in\mathbb{Z}^d}$ are **identically distributed** if there exists a Borel probability measure μ on $\mathbb{R}$ such that $\mathbb{P}(\omega_n \in A) = \mu(A)$ for all $n \in \mathbb{Z}^d$ and Borel sets $A \subset \mathbb{R}$.
$(\omega_n)_{n\in\mathbb{Z}^d}$ are **independent** if

$$
\mathbb{P}(\omega_{n_1} \in A_1, \ldots, \omega_{n_l} \in A_l) = \prod_{k=1}^{l} \mathbb{P}(\omega_{n_k} \in A_k) = \prod_{k=1}^{l} \mu(A_k)
$$

for each finite subset $\{n_1, \ldots, n_l\}$ of $\mathbb{Z}^d$ and arbitrary Borel sets $A_1, \ldots, A_l \subset \mathbb{R}$.

In this thesis we assume that ω_n are uniformly distributed in the real closed interval $[-1, 1]$. This means that $\mu(\omega_n) = \rho(\omega_n)d\omega_n$ where the continuous density is given by $\rho = \frac{1}{2}\chi_{[-1,1]}$.
It is possible to generalize the results to all absolutely continuous distributions where ρ has compact support and to the Gaussian distribution.

Physically, V_ω represents the random electric potential created by nuclei at the sites $n \in \mathbb{Z}^d$.
As a diagonal operator, the spectrum of V_ω consists of the random entries ω_n, each corresponding to an eigenvector given by the Dirac delta $\delta_n \in \ell^2(\mathbb{Z}^d)$.

As for the descrete laplacian, the random term is summable in $\ell^2(\mathbb{Z}^d)$, bounded and self-adjoint. Self-adjointness of the maximal multiplication operator by a real-valued function is a fact. The spectrum is the closed interval $[-1, 1] \subset \mathbb{R}$, which means that V_ω is bounded, and since $\|V_\omega u\|^2_{\ell^2} \leq \|u\|^2_{\ell^2}$ it is also summable in $\ell^2(\mathbb{Z}^d)$.

The Disorder Parameter is represented by $\lambda > 0$. The value of λ is a great indicator for localisation, which makes it one of the central objects of interest of this thesis. Let's consider the two extreme cases:

- For $\boldsymbol{\lambda = 0}$ the Hamiltonian looses the random term and only considers the Laplacian. Without disorder, the material behaves like a perfect crystal without localisation. The state functions are continuous plane waves and the spectrum is also continuous between $[0, 4d]$.
- For $\boldsymbol{\lambda \gg 1}$ the kinetic hopping term becomes negligible compared to the random potential, so that $H_\omega \approx \lambda V_\omega$. In this scenario electrons are unable to hop between adjacent sites, and the material behaves as an insulator. In this limit, the system exhibits trivial localization: the eigenfunctions are localized Dirac deltas δ_x, and the spectrum is confined to the compact interval $[-\lambda, \lambda]$

2.2 The spectrum

- in $\mathbb{R}$
- bounded $[0, 4d] + \text{supp}\mu$
- spectral division (any use of the spectral theorem should be sent to the Appendix)
- RAGE theorem (or maybe directly in the Appendix or chapter 2)

Dalla teoria spettrale per operatori autoaggiunti su uno spazio di Hilbert $\mathcal{H}$, lo spazio si decompone univocamente nella somma diretta di sottospazi invarianti ortogonali:

$$\mathcal{H} = \mathcal{H}_{pp} \oplus \mathcal{H}_{sc} \oplus \mathcal{H}_{ac} \tag{2.5}$$

a cui corrispondono rispettivamente lo spettro puramente puntuale $\sigma_{pp}(H)$, singolarmente continuo $\sigma_{sc}(H)$ e assolutamente continuo $\sigma_{ac}(H)$, con $\sigma(H) = \sigma_{pp}(H) \cup \sigma_{sc}(H) \cup \sigma_{ac}(H)$.

L'interpretazione fisica di questi sottospazi è chiarita dal fondamentale **Teorema RAGE** (Ruelle-Amrein-Georgescu-Enss):

Theorem 2.2 (RAGE). *Suppose H is a Schrödinger operator acting on $\ell^2(\mathbb{Z}^d)$. Then:*

(i) $\phi \in \mathcal{H}_{pp}$ *if and only if*

$$\lim_{R\to\infty} \sup_{t\in\mathbb{R}} \|\chi_{(|x|>R)} e^{-itH}\phi\| = 0.$$

(ii) $\psi \in \mathcal{H}_c = \mathcal{H}_{ac} \oplus \mathcal{H}_{sc}$ *if and only if*

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \|\chi_{(|x|\leq R)} e^{-itH}\psi\| \, dt = 0$$

for all $R > 0$.

Here $\chi_{(|x|\leq R)}$ and $\chi_{(|x|>R)}$ denote the characteristic functions of the ball $\{|x| \leq R\}$ and the complement of the ball, respectively.

2.3 Main results

In this section one could write first of all the main theorem and the spectral results. starting maybe with:
The aim of this thesis is to prove localisation on the whole spectrum of H_ω ...
Poi scrivo il teorema finale con spiegazione che la dimostrazione avverrà nel capitolo 5.
potrebbe essere un'idea anche scrivere una piccola parentesi sulla differenza dei rislutati con Frohlich e spencer e con Aizenman e Molchanov.
One of the main results concerns the values of λ

- $\lambda >> 1$ and $\lambda = 0$
- general case for $d = 1, d \geq 2$

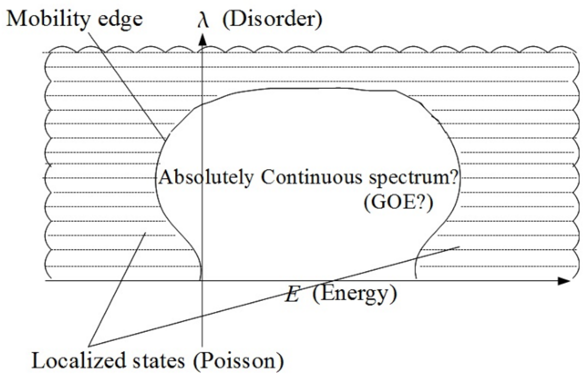

Figure 1: Aizenman figure

3 Localisation properties

We understood what localisation means from an heuristical point of view. Mathematically, it can be described from spectral and dynamical properties.

3.1 spectral localisation

Definition 3.1 (Spectral Localization)**.** random Schrödinger operator $H = H_\omega$ exhibits **spectral localization** if H_ω almost surely has only pure point spectrum, that is,

$$\sigma(H_\omega) = \sigma_{pp}(H_\omega) \quad \text{and} \quad \sigma_c(H_\omega) = \emptyset \quad \text{almost surely.}$$

Furthermore the eigenfunctions corresponding to all eigenvalues decay exponentially: for almost all ω, H_ω has a complete set of eigenfunctions $(\phi_{\omega,n})_{n\in\mathbb{N}}$ satisfying

$$|\phi_{\omega,n}(x)| \le C_{\omega,n} e^{-\mu|x-x_{n,\omega}|}$$

where $\mu > 0$ is the localization length parameter, $x_{n,\omega}$ are the centers of localization, and $C_{\omega,n} < \infty$ is a finite constant.

3.1.1 Absence of transport

$$\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t) = \left\| \, |x| \, e^{-itH_\omega} P_{H_\omega \in [a,b]}\phi_0 \right\|^2 \tag{5}$$

If the electrons with energies in $[a, b]$ move ballistically with average velocity v_{av}, then roughly $x(t) \sim v_{av}t$ for large times. Hence $\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t)$ will be proportional to t^2 for large t.

On the other hand, if the electrons in the energy interval $[a, b]$ are localized, then it should be natural to expect that $\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t)$ is bounded uniformly in t. It is known (Simon, 1990) that pure point spectrum implies absence of ballistic motion,

$$\lim_{t\to\infty} \frac{\langle x^2 \rangle_{P_{H_\omega \in [a,b]}\phi_0}(t)}{t^2} = 0$$

for all compactly supported initial conditions ϕ_0, as soon as the spectrum in $[a, b]$ is pure point. However, del Rio et al. (1996) constructed examples of (non-random) one-dimensional Schrödinger operators with pure point spectrum for all energies and exponentially localized eigenfunctions for which

$$\limsup_{t\to\pm\infty} \frac{\langle x^2\rangle_{\phi_0}(t)}{t^{2-\delta}} = \infty \qquad \text{for all } \delta > 0, \tag{6}$$

for a large class of compactly supported initial conditions ϕ_0. *Il fallimento risiede nella libertà delle costanti $C_{\omega,n}$: se $C_{\omega,n}$ cresce incontrollatamente con n, le code delle autofunzioni possono sovrapporsi su scale spaziali arbitrariamente grandi.*

3.2 dynamical localisation

Definition 3.2 (Dynamical localisation)**.**

$$\sup_{t\in\mathbb{R}} \||X|^p e^{-itH_\omega}\psi\| < \infty \quad \textit{almost surely,} \tag{3.1}$$

where the position operator $|X|$ is defined by $(|X|\phi)(n) = |n|\phi(n)$ and for all φ with compact support..

Definition 3.3 (Exponential localisation)**.**

$$\mathbb{E}\left(\sup_{t\in\mathbb{R}} |\langle e_j, e^{-itH_\omega} e_k\rangle|\right) \le Ce^{-\mu|j-k|} \tag{3.2}$$

Remark 3.4. Dynamical localization in the form (2.2) is a strong form of asserting that solutions of the time-dependent Schrödinger equation $H_\omega\phi(t) = i\partial\phi(t)$ are staying localized in space, uniformly for all times, and thus shows the absence of quantum transport.

Proposition: *Exponential localization implies that all moments of the position operator are bounded in time, i.e., dynamical localisation*

Proof. To see how (2.1) follows from (2.2), we assume that $\psi(k) = 0$ for $|k| > R$. Then

$$\begin{aligned}
\||X|^p e^{-ith_\omega}\psi\|^2 &= \sum_j |\langle e_j, |X|^p e^{-ith_\omega}\psi\rangle|^2 \\
&= \sum_j |j|^{2p} \left|\sum_{|k|\le R} \langle e_j, e^{-ith_\omega} e_k\rangle \psi(k)\right|^2 \\
&\le \sum_j \sum_{|k|\le R} |j|^{2p} |\langle e_j, e^{-ith_\omega} e_k\rangle|^2 \|\psi\|^2,
\end{aligned}$$

where the last step used the Cauchy-Schwarz inequality. We can drop the square from $|\langle e_j, e^{-ith_\omega} e_k\rangle|^2$ (as this number is bounded by 1) and then take expectations to get

$$\begin{aligned}\mathbb{E}\left(\sup_t \||X|^p e^{-ith_\omega}\psi\|^2\right) &\le \sum_j \sum_{|k|\le R} |j|^{2p}\mathbb{E}\left(\sup_t |\langle e_j, e^{-ith_\omega} e_k\rangle|\right)\|\psi\|^2 \\ &\le C\sum_j \sum_{|k|\le R} |j|^{2p} e^{-\mu|j-k|}\|\psi\|^2 \\ &< \infty.\end{aligned}$$

□

3.3 From dynamical to spectral localisation

Proposition 3.5. *Suppose that dynamical localization in the form*

$$\mathbb{E}\left(\sup_{t\in\mathbb{R}} |\langle e_j, e^{-ith_\omega}\chi_I(h_\omega)e_k\rangle|\right) \le Ce^{-\mu|j-k|} \tag{3.3}$$

holds in an open interval I. Then h_ω almost surely has pure point spectrum in I.

Proof. For a discrete Schrödinger operator $h = h_0 + V$ in $\ell^2(\mathbb{Z}^d)$ let $P_{\text{cont}}(h)$ be the projection onto its continuous spectral subspace. Then the RAGE-Theorem says that for every $\psi \in \ell^2(\mathbb{Z}^d)$,

$$\|P_{\text{cont}}(h)\chi_I(h)\psi\|^2 = \lim_{R\to\infty}\lim_{T\to\infty}\int_0^T \frac{dt}{T}\|\chi_{\{|x|\ge R\}}e^{-ith}\chi_I(h)\psi\|^2. \tag{3.4}$$

If ψ has finite support, say $\operatorname{supp}\psi \subset \{|x| \le r\}$, then

$$\begin{aligned}\|\chi_{\{|x|\ge R\}}e^{-ith}\chi_I(h)\psi\|^2 &\le \|\chi_{\{|x|\ge R\}}e^{-ith}\chi_I(h)\chi_{\{|x|\le r\}}\|^2\|\psi\|^2 \\ &\le \|\chi_{\{|x|\ge R\}}e^{-ith}\chi_I(h)\chi_{\{|x|\le r\}}\|\|\psi\|^2 \\ &\le \sum_{|x|\ge R, |y|\le r} |\langle e_x, e^{-ith}\chi_I(h)e_y\rangle|\|\psi\|^2,\end{aligned}$$

where dropping a square is allowed as $\|\chi_{\{|x|\ge R\}}e^{-ith}\chi_I(h)\chi_{\{|x|\le r\}}\| \le 1$.

Taking expectations in (2) implies, after using Fatou and Fubini,

$$\mathbb{E}\left(\|P_{\mathrm{cont}}(h_\omega)\chi_I(h_\omega)\psi\|^2\right) \leq \lim_{R\to\infty, T\to\infty} \int_0^T \frac{dt}{T} \sum_{|x|\geq R, |y|\leq r} \mathbb{E}\left(|\langle e_x, e^{-ith_\omega}\chi_I(h_\omega)e_y\rangle|\right) \|\psi\|^2. \tag{3.5}$$

By (2.4) we have $\mathbb{E}\left(|\langle e_x, e^{-ith_\omega}\chi_I(h_\omega)e_y\rangle|\right) \leq Ce^{-\mu|x-y|}$ uniformly in t, which bounds the right hand side of (2.6) by

$$\leq \lim_{R\to\infty} \tilde{C} \sum_{|x|\geq R, |y|\leq r} e^{-\mu|x-y|} = 0.$$

We conclude that $P_{\mathrm{cont}}(h_\omega)\chi_I(h_\omega)\psi = 0$ for almost every ω and every ψ of finite support. The latter are dense in $\ell^2(\mathbb{Z}^d)$ and thus $P_{\mathrm{cont}}(h_\omega)\chi_I(h_\omega) = 0$ almost surely, meaning that the spectrum in I is pure point. $\square$

4 Fractional moments

In this section we are going to see the important lemmas and theorems that help us proving localisation. In particular we see the power of the fractional moment method.

4.1 The green function

The *resolvent operator* is defined as:

$$G(z) = (H - z)^{-1}$$

4.2 Localisation at large disorder

Theorem 4.1 (Aizenman–Molchanov, 1993)**.** *Let $0 < s < 1$. Then there exists $\lambda_0 > 0$ such that for $\lambda \geq \lambda_0$ there are $C < \infty$ and $\mu > 0$ with*

$$\mathbb{E}\left(|G_{\omega,\lambda}(x, y; z)|^s\right) \leq Ce^{-\mu|x-y|} \tag{4.1}$$

uniformly in $x, y \in \mathbb{Z}^d$ and $z \in \mathbb{C} \setminus \mathbb{R}$. In particular, if λ_{And} satisfies $\lambda_{\text{And}} = \mu_d e \ln \lambda_{And}$ then for all $\lambda > \lambda_{\text{And}}, \lambda > e$ and $\epsilon > 0$ we have

$$\mathbb{E}\left(|G^{(\Lambda)}(x, y; z)|^{1-\frac{1}{\ln \lambda}}\right) \leq \beta(\lambda) K_\epsilon \ln \lambda \, e^{-m_\epsilon(\lambda)|x-y|} \tag{4.2}$$

where

$$m_\epsilon(\lambda) = -\ln \beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

Lemma 4.2 (A priori bound)**.** *There exists a constant $C_1 = C_1(s, \rho) < \infty$ such that*

$$\mathbb{E}_x(|G_{\omega,\lambda}(x, x; z)|^s) \leq C_1 \lambda^{-s} \tag{4.3}$$

for all $x \in \mathbb{Z}^d$, $z \in \mathbb{C} \setminus \mathbb{R}$, and $\lambda > 0$.

Here

$$C_1 = \frac{1}{1 - s}$$

for $\omega_x \sim \mathcal{U}[-1, 1]$

Lemma 4.3. ***RESOLVENT IDENTITY*** *còmo en el lìbro*

Proof of a priori bound. This proof demonstrates the simplicity of the fundamental idea underlying the FMM. For fixed $x \in \mathbb{Z}^d$, write $\omega = (\hat{\omega}, \omega_x)$ where $\hat{\omega}$ is short for $(\omega_u)_{u \in \mathbb{Z}^d \setminus \{x\}}$. With $P_{e_x} := \langle e_x, \cdot \rangle e_x$, the orthogonal projection onto the span of e_x, we can separate the ω_x and $\hat{\omega}$ dependence of H_ω

as

$$H_\omega = -\Delta + \lambda V_{\hat\omega} + \lambda\omega_x P_{e_x} = H_{\hat\omega} + \lambda\omega_x P_{e_x}.$$

For the operators $(H_\omega - z)$ and $(H_{\hat\omega} - z)$, the resolvent identity yields

$$(H_\omega - z)^{-1} = (H_{\hat\omega} - z)^{-1} - \lambda\omega_x (H_{\hat\omega} - z)^{-1} P_{e_x} (H_\omega - z)^{-1}. \tag{4.4}$$

Taking matrix-elements we conclude for the corresponding diagonal Green functions that

$$G_{\omega,\lambda}(x,x;z) = G_{\hat\omega,\lambda}(x,x;z) - \lambda\omega_x G_{\hat\omega,\lambda}(x,x;z) G_{\omega,\lambda}(x,x;z) \tag{4.5}$$

and

$$G_{\omega,\lambda}(x,x;z) = \frac{1}{a+\lambda\omega_x} \quad \text{with } a = \frac{1}{G_{\hat\omega,\lambda}(x,x;z)}. \tag{4.6}$$

Note that the latter is well-defined since one can easily check that $\operatorname{Im} G_{\hat\omega,\lambda}(x,x;z) > 0$ for $\operatorname{Im} z > 0$.

The important fact is that a is a complex number which does not depend on ω_x. Thus, writing $\mathbb{E}_x(\dots) := \int \dots \rho(\omega_x)\, d\omega_x$ for $\rho(\omega_x) = \frac{1}{2}\chi_{[-1,1]}$, we find that

$$\mathbb{E}_x(|G_{\omega,\lambda}(x,x;z)|^s) = \frac{1}{2\lambda^s}\int_{-1}^{1} \frac{d\omega_x}{|\frac{a}{\lambda} + \omega_x|^s} \le \frac{1}{2\lambda^s}\int_{-1}^{1} \frac{d\omega_x}{|\omega_x|^s} = \frac{1}{(1-s)\lambda^s}, \tag{4.7}$$

with $\frac{1}{1-s}$ independent of λ and a, and thus independent of $\hat\omega$, z and x.

□

4.2.1 From fractional moment's exponential bound to dynamical localisation

In this subsection we aim to prove the following theorem:

Theorem 4.4. *Let $I \subset \mathbb{R}$ be an open bounded interval. If there exist $s \in (0,1)$, $C < \infty$ and $\mu > 0$ such that*

$$\mathbb{E}\left(|G(x,y;E+i\eta)|^s\right) \le Ce^{-\mu|x-y|} \tag{4.8}$$

uniformly in $E \in I$ and $\eta > 0$, then dynamical localisation holds on the interval I.

Before diving in the proof, we are going to need more equipment.

Lemma 4.5. *For a bounded and self-adjoint operator A,*

$$\left\|(A-z)^{-1}\right\| \leq \frac{1}{\mathrm{Im}(z)} \tag{4.9}$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$.

Proof. For the definition of the operator norm, we know that

$$\left\|(A-z)^{-1}\right\| := \sup_{\mu \in \sigma((A-z)^{-1})} |\mu|$$

Let's consider a $\lambda \in \mathbb{C}$ and write $\mu = \frac{1}{\lambda - z}$. This means:

$$\begin{aligned}
\frac{1}{\lambda - z} \in \sigma((A-z)^{-1}) &\iff (A-z)^{-1} - (\lambda - z)^{-1} \text{ not invertible} \\
&\iff (A-z)^{-1}(A-\lambda)(\lambda-z)^{-1} \text{ not invertible} \\
&\iff (A-\lambda) \text{ not invertible} \\
&\iff \lambda \in \sigma(A)
\end{aligned}$$

Where the resolvent identity has been used in third Implication.
At this point we obtain

$$\left\|(A-z)^{-1}\right\| = \sup_{\lambda \in \sigma(A)} \frac{1}{|\lambda - z|} = \frac{1}{\inf_{\lambda \in \sigma(A)} |\lambda - z|} = \frac{1}{\mathrm{dist}(\sigma(A), z)}$$

For a bounded and self-adjoint operator, we know that the spectrum is real-valued. For $z \in \mathbb{C} \setminus \mathbb{R}$ the minimum distance between z and $\sigma(A)$ is the imaginary component of z:

$$\left\|(A-z)^{-1}\right\| \leq \frac{1}{|\mathrm{Im}(z)|} \tag{4.10}$$

□

Thid bound is one of the most important and useful in this thesis. Once again we applied the resolvent identity to find a simple yet useful result.

Proposition 4.6. *For every $s \in (0,1)$ there exists a constant $C_1 < \infty$ only depending on s and ρ such that*

$$|\mathrm{Im}\, z|\, \mathbb{E}_x\left(|G(x,y;z)|^2\right) \leq C_1 \mathbb{E}_x\left(|G(x,y;z)|^s\right) \tag{4.11}$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$ and $x, y \in \mathbb{Z}^d$.

Proof. As in the proof of Lemma 3.2 we write $\omega = (\hat{\omega}, \omega_x)$. Keep $\hat{\omega}$ fixed and consider the Hamiltonian

$$H^{(\alpha)} = H(\hat{\omega}, \omega_x + \alpha) = H_\omega + \alpha P_{e_x}.$$

Its Green function will be denoted by $G^{(\alpha)}$. Applying the resolvent identity, we find

$$(H_\omega - z)^{-1} = (H^{(\alpha)} - z)^{-1} + \alpha (H_\omega - z)^{-1} P_{e_x} (H^{(\alpha)} - z)^{-1}$$

and

$$\begin{aligned} G^{(\alpha)}(x, y; z) &= \frac{G_\omega(x, y; z)}{1 + \alpha G_\omega(x, x; z)} \\ &= \frac{1}{\alpha + G_\omega(x, x; z)^{-1}} \cdot \frac{G_\omega(x, y; z)}{G_\omega(x, x; z)}. \end{aligned} \tag{4.12}$$

For the special case $x = y$ and $\tilde{\alpha} = -\operatorname{Re} G_\omega(x, x; z)^{-1}$ we get from (3.11) and Lemma 3.5 that

$$\left| \frac{1}{\operatorname{Im} G(x, x; z)^{-1}} \right| = |G^{(\tilde{\alpha})}(x, x; z)| \leq \frac{1}{|\operatorname{Im} z|},$$

i.e. $|\operatorname{Im} G(x, x; z)^{-1}| \geq |\operatorname{Im} z|$. Inserting this into (3.11) gives

$$|\operatorname{Im} z| \, |G^{(\alpha)}(x, y; z)|^2 \leq \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2} \cdot \frac{|G_\omega(x, y; z)|^2}{|G_\omega(x, x; z)|^2}. \tag{4.13}$$

On the other hand, we can bound the same expression by

$$\begin{aligned} |\operatorname{Im} z| \, |G^{(\alpha)}(x, y; z)|^2 &\leq |\operatorname{Im} z| \sum_{y' \in \mathbb{Z}^d} |G^{(\alpha)}(x, y'; z)|^2 \\ &= |\operatorname{Im} z| \langle e_x \, (H^{(\alpha)} - z)^{-1}, (H^{(\alpha)} - z)^{-1} \, e_x \rangle \\ &= |\operatorname{Im} z| \langle e_x, (H^{(\alpha)} - \bar{z})^{-1} (H^{(\alpha)} - z)^{-1} e_x \rangle \\ &= |\operatorname{Im} z| \langle e_x, \frac{1}{z - \bar{z}} \left[(H^{(\alpha)} - z)^{-1} - (H^{(\alpha)} - \bar{z})^{-1} \right] e_x \rangle \\ &= |\operatorname{Im} G^{(\alpha)}(x, x; z)| \\ &= \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2}, \end{aligned} \tag{4.14}$$

where the third step used the symmetry of $H^{(\alpha)}$ and last step used (3.11) with $x = y$.

For $t \geq 0$ one has $\min(1, t^2) \leq t^s$, with $s \in (0,1)$. Using this to interpolate between (3.12) and (3.13) we get

$$|\mathrm{Im}\, z|\, |G^{(\alpha)}(x,y;z)|^2 \leq \frac{|\mathrm{Im}\, G_\omega(x,x;z)^{-1}|}{|\alpha + G_\omega(x,x;z)^{-1}|^2} \cdot \frac{|G_\omega(x,y;z)|^s}{|G_\omega(x,x;z)|^s}. \tag{4.15}$$

We will now use the following "re-sampling trick", which has the effect of creating an additional random variable (here α) to average over. For a non-negative Borel function f on $\mathbb{R}$,

$$\begin{aligned}\iint f(\omega_x+\alpha)\rho(\omega_x+\alpha)\, d\alpha\, \rho(\omega_x)\, d\omega_x &= \iint f(\omega_x+\alpha)\rho(\omega_x+\alpha)\rho(\omega_x)\, d\omega_x\, d\alpha \\ &= \iint f(\omega_x)\rho(\omega_x)\rho(\omega_x-\alpha)\, d\omega_x\, d\alpha \\ &= \int f(\omega_x)\rho(\omega_x)\left(\int \rho(\omega_x-\alpha)\, d\alpha\right) d\omega_x \\ &= \int f(\omega_x)\rho(\omega_x)\, d\omega_x, \end{aligned} \tag{4.16}$$

where in the first and third steps we used Fubini and in the second we used translation invariance of Lebesgue measure.

Choose $f(\omega_x) = |G_{(\hat{\omega},\omega_x)}(x,y;z)|^2$; then (3.14) and (3.15) yield

$$\begin{aligned}|\mathrm{Im}\, z|\, \mathbb{E}_x(|G_\omega(x,y;z)|^2) &= |\mathrm{Im}\, z|\, \mathbb{E}_x\left(\int |G^{(\alpha)}(x,y;z)|^2 \rho(\omega_x+\alpha)\, d\alpha\right) \\ \leq \mathbb{E}_x&\left(|\mathrm{Im}\, G_\omega(x,x;z)^{-1}|\frac{|G_\omega(x,y;z)|^s}{|G_\omega(x,x;z)|^s}\int \frac{\rho(\omega_x+\alpha)}{|\alpha+G_\omega(x,x;z)^{-1}|^2}\, d\alpha\right).\end{aligned} \tag{4.17}$$

We now use Lemma 3.7 below with $w = G_\omega(x,x;z)^{-1}$ to conclude

$$|\mathrm{Im}\, z|\, \mathbb{E}_x(|G_\omega(x,y;z)|^2) \leq C\mathbb{E}_x(|G_\omega(x,y;z)|^s),$$

with a constant $C < \infty$ which only depends on $\mathrm{supp}\, \rho$, but not on x, y and z. $\square$

Lemma 4.7. *Let $0 < s < 1$. There exists a constant $C_s' < \infty$ such that*

$$|\mathrm{Im}\, w| \cdot |w|^s \int \frac{\rho(\omega_x+\alpha)}{|\alpha+w|^2}\, d\alpha \leq C_s'$$

uniformly in $w \in \mathbb{C}$ and $\omega_x \sim \mathcal{U}[-1,1]$.

Proof. The proof is that of [REFERENCE GUNTER STOLZ]. Since $|w|^s \leq |\alpha|^s + |\alpha + w|^s$, we bound:

1.

$$|\mathrm{Im}\, w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2}\, d\alpha \leq |\mathrm{Im}\, w| \int \frac{\| |\alpha|^s \rho(\omega_x + \alpha) \|_\infty}{|\alpha + w|^2}\, d\alpha$$
$$\leq |\mathrm{Im}\, w|\, \| |\alpha|^s \rho(\omega_x + \alpha) \|_\infty \int \frac{1}{(\alpha + \mathrm{Re}\, w)^2 + \mathrm{Im}\, w^2}\, d\alpha$$

for $a, b \in \mathbb{R}$ we have

$$\int \frac{1}{a^2 + b^2}\, da = 2\pi i \,\mathrm{Res}\left(\frac{1}{a^2+b^2}, ib\right) = \frac{2\pi i}{2ib}.$$

Thus for $\rho = \frac{1}{2}\chi_{[-1,1]}$ and $|\alpha| \leq 1 + |\omega| \leq 2$

$$|\mathrm{Im}\, w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2}\, d\alpha \leq \pi \| |\alpha|^s \rho(\omega_x + \alpha) \|_\infty$$
$$\leq \pi 2^{s-1}.$$

2. Now, we bound the second expression.

   (a) For $|\mathrm{Im}\, w|$ big:
   Since $|\alpha + w| \geq |\,\mathrm{Im}\, w|$ and $\int \rho(\omega_x + \alpha)\, d\alpha = 1$:

   $$I_2 \leq |\,\mathrm{Im}\, w| \cdot \frac{1}{|\,\mathrm{Im}\, w|^{2-s}} \int \rho(\omega_x + \alpha)\, d\alpha = \frac{1}{|\,\mathrm{Im}\, w|^{1-s}}$$

   (b) For $|\mathrm{Im}\, w|$ small:

   $$|\mathrm{Im}\, w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}}\, d\alpha \leq |\mathrm{Im}\, w| \frac{1}{2} \int \frac{d\alpha}{|\alpha + w|^{2-s}}$$
   $$= |\mathrm{Im}\, w| \frac{1}{2} \int \frac{1}{|u + \mathrm{Im} w|^{2-s}}\, du$$
   $$= |\mathrm{Im}\, w|^2 \frac{1}{2} \int \frac{1}{|u^2 + \mathrm{Im} w^2|^{\frac{2-s}{2}}}\, du$$
   $$= \frac{1}{2} (\mathrm{Im}\, w)^s \int \frac{1}{|t^2 + 1|^{\frac{2-s}{2}}}\, dt$$

by setting $u = \alpha + \operatorname{Re} w$ and $t = \frac{u}{\operatorname{Im} w}$.

To establish the finiteness of the constant, we consider the integral

$$C_s := \int_{-\infty}^{\infty} \frac{1}{(t^2+1)^{\frac{2-s}{2}}}\, dt.$$

For $s \in (0,1)$, the exponent satisfies $2-s > 1$, which ensures the asymptotic decay $(t^2+1)^{-\frac{2-s}{2}} \sim |t|^{-(2-s)}$ as $|t| \to \infty$, rendering the integral absolutely convergent.Then we have

$$|\operatorname{Im} w| \int \frac{\rho(\omega_x+\alpha)}{|\alpha+w|^{2-s}}\, d\alpha \leq \min\left(\frac{1}{|\operatorname{Im} w|^{1-s}}, C_s \frac{1}{2}|\operatorname{Im} w|^s\right) \leq C_s' 2^{s-1}.$$

$\square$

Now we can procede with the proof of the theorem.

Proof of Theorem 3.4. Let $\mu_{x,y}$ be the complex Borel spectral measure of h_ω associated with vectors δ_x, δ_y, defined by

$$\mu_{x,y}(B) := \langle \delta_x, \chi_B(h_\omega)\delta_y\rangle.$$

The total variation measure $|\mu_{x,y}|$ over the open bounded interval I satisfies

$$\sup_{t\in\mathbb{R}} |\langle \delta_x, e^{-ith_\omega}\chi_I(h_\omega)\delta_y\rangle| \leq |\mu_{x,y}|(I).$$

By Lusin's Theorem, $|\mu_{x,y}|(I) = \sup_{g\in C_c(I), \|g\|_\infty \leq 1} |\langle \delta_x, g(h_\omega)\delta_y\rangle|$. For every $g \in C_c(I)$, the spectral calculus and Parseval's identity yield

$$\langle \delta_x, g(h_\omega)\delta_y\rangle = \lim_{\eta\downarrow 0} \frac{\eta}{\pi} \int_I g(E)\langle \delta_x, (h_\omega - E - i\eta)^{-1}(h_\omega - E + i\eta)^{-1}\delta_y\rangle\, dE.$$

Inserting a partition of unity I:

$$|\langle \delta_x, g(h_\omega)\delta_y\rangle| \leq \liminf_{\eta\downarrow 0} \frac{\eta}{\pi} \int_I \sum_{z\in\mathbb{Z}^d} |G_\omega(x,z;E+i\eta)|\,|G_\omega(z,y;E+i\eta)|\, dE.$$

Taking expectations, applying Fatou's Lemma, Fubini's Theorem, and the

Cauchy–Schwarz inequality on the expectation gives:

$$\begin{aligned}\mathbb{E}(|\mu_{x,y}|(I)) &\leq \frac{1}{\pi}\liminf_{\eta\downarrow 0}\int_I \sum_{z\in\mathbb{Z}^d}\mathbb{E}\Big(\eta|G_\omega(x,z;E+i\eta)|\,|G_\omega(z,y;E+i\eta)|\Big)\,dE \\ &\leq \frac{1}{\pi}\liminf_{\eta\downarrow 0}\int_I \sum_{z\in\mathbb{Z}^d}\big(\mathbb{E}\big(\eta|G_\omega(x,z;E+i\eta)|^2\big)\big)^{1/2}\big(\mathbb{E}\big(\eta|G_\omega(z,y;E+i\eta)|^2\big)\big)^{1/2}\,dE.\end{aligned}$$

Applying Proposition to both factors and then substituting the uniform hypothesis:

$$\begin{aligned}\mathbb{E}(|\mu_{x,y}|(I)) &\leq \frac{C_1}{\pi}\int_I \sum_{z\in\mathbb{Z}^d}(\mathbb{E}|G_\omega(x,z;E+i\eta)|^s)^{1/2}(\mathbb{E}|G_\omega(z,y;E-i\eta)|^s)^{1/2}\,dE \\ &\leq \frac{C_1|I|C}{\pi}\sum_{z\in\mathbb{Z}^d} e^{-\frac{\mu}{2}|x-z|}e^{-\frac{\mu}{2}|z-y|}.\end{aligned}$$

By the triangle inequality, $\frac{\mu}{2}|x-z|+\frac{\mu}{2}|z-y| \geq \frac{\mu}{4}|x-y|+\frac{\mu}{4}|x-z|+\frac{\mu}{4}|z-y|$. Therefore,

$$\mathbb{E}(|\mu_{x,y}|(I)) \leq \left(\frac{C_1|I|C}{\pi}\sum_{z\in\mathbb{Z}^d} e^{-\frac{\mu}{4}|x-z|}e^{-\frac{\mu}{4}|z-y|}\right)e^{-\frac{\mu}{4}|x-y|} \leq \widetilde{C}e^{-\widetilde{\mu}|x-y|},$$

with $\widetilde{\mu} = \frac{\mu}{4}$ and finite prefactor $\widetilde{C} = \frac{C_1|I|C}{\pi}\sum_{u\in\mathbb{Z}^d} e^{-\frac{\mu}{4}|u|}$. □

Remark 4.8 (Spectral Localization). Via the RAGE theorem, spectral localisation is also implied.

5 SAW representation

5.1 SAW Definitions

Definition 5.1 (Self-Avoiding Walk)**.** A **self-avoiding walk** (SAW) on the integer lattice $\mathbb{Z}^d$ of length $N \in \mathbb{N}_0$ is an ordered $(N+1)$-tuple of distinct lattice sites

$$\gamma = (\gamma_0, \gamma_1, \dots, \gamma_N) \in (\mathbb{Z}^d)^{N+1}$$

such that consecutive vertices are nearest neighbors, that is,

$$|\gamma_i - \gamma_{i-1}| = 1 \quad \text{for all } i \in \{1, \dots, N\},$$

and $\gamma_i \neq \gamma_j$ for all $i \neq j$.

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Definition 5.2 (Sets of Self-Avoiding Walks)**.** Let $\Lambda \subseteq \mathbb{Z}^d$. For any $x, y \in \Lambda$ and $N \in \mathbb{N}_0$:

(i) $\mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N contained entirely in Λ, starting at $\gamma_0 = x$ and terminating at $\gamma_N = y$. When $\Lambda = \mathbb{Z}^d$, we simply write $\mathcal{S}_N(y, x)$.

(ii) $\mathcal{S}_N^{(\Lambda)}(x) := \bigcup_{y \in \Lambda} \mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N in Λ starting at x.

(iii) $c_N := |\mathcal{S}_N(x)| = \sum_{y \in \mathbb{Z}^d} |\mathcal{S}_N(y, x)|$ denotes the total number of self-avoiding walks of length N on $\mathbb{Z}^d$ originating from a fixed site (e.g., the origin 0), which is independent of the starting vertex by translation invariance.

Definition 5.3 (Connectivity Constant)**.** The **connective constant** (or connectivity constant) μ_d of the lattice $\mathbb{Z}^d$ is defined as the asymptotic growth rate of c_N:

$$\mu_d := \lim_{N \to \infty} (c_N)^{1/N}.$$

By subadditivity of $\log c_N$, this limit exists, is positive, and satisfies the strict inequality $\mu_d < 2d - 1$.

Claim 5.4. Let μ_d be the connective constant (or growth rate) of self-avoiding walks on $\mathbb{Z}^d$. Then:

$$\mu_d \leq 2d - 1$$

Proof. Let c_N denote the number of self-avoiding walks (SAWs) of length N starting from the origin $0 \in \mathbb{Z}^d$, and let $S_N(0,x)$ denote the set of such walks ending at $x \in \mathbb{Z}^d$. By definition:

$$c_N = \sum_{x\in\mathbb{Z}^d} |S_N(0,x)| = \sum_{x\in\mathbb{Z}^d} \sum_{\gamma\in S_N(0,x)} 1$$

Consider the generating function (*susceptibility*) with parameter $\xi \in \mathbb{C}$:

$$\sum_{N=0}^{\infty} |\xi|^N c_N = \sum_{N=0}^{\infty} |\xi|^N \sum_{x\in\mathbb{Z}^d} |S_N(0,x)|$$

For $N = 0$, $c_0 = 1$. For $N \geq 1$, a walk cannot immediately step back onto the previous vertex (non-reversing condition). Thus:

- The first step has at most $2d$ choices.
- Each of the subsequent $N-1$ steps has at most $2d-1$ choices.

This gives the upper bound:

$$c_N \leq 2d(2d-1)^{N-1}, \quad \text{for } N \geq 1.$$

Using this upper bound, we can bound the series from above:

$$\begin{aligned}
\sum_{N=0}^{\infty} |\xi|^N c_N &\leq 1 + \sum_{N=1}^{\infty} |\xi|^N (2d)(2d-1)^{N-1} \\
&= 1 + 2d|\xi| \sum_{k=0}^{\infty} |\xi|^k (2d-1)^k \quad (\text{setting } k = N-1) \\
&= 1 + 2d|\xi| \sum_{k=0}^{\infty} (|\xi|(2d-1))^k
\end{aligned}$$

This geometric series converges if and only if:

$$|\xi|(2d-1) < 1 \iff |\xi| < \frac{1}{2d-1} =: R$$

where $R = \frac{1}{2d-1}$ is the radius of convergence of the bounding series.

By the Cauchy–Hadamard theorem, the radius of convergence R' of the original series $\sum_{N=0}^{\infty} c_N |\xi|^N$ is:

$$R' = \frac{1}{\limsup_{N\to\infty} c_N^{1/N}} = \frac{1}{\mu_d}$$

Since the original series is bounded term-by-term by a series that converges for $|\xi| < R$, the true radius of convergence must satisfy:

$$R' \geq R \implies \frac{1}{\mu_d} \geq \frac{1}{2d-1}$$

Taking the reciprocal yields:

$$\mu_d \leq 2d - 1.$$

$\square$

Definition 5.5 (SAW Correlation Function)**.** For a parameter $\beta \geq 0$, the **self-avoiding walk correlation function** $C_\beta(y-x)$ between two vertices $x, y \in \mathbb{Z}^d$ is the generating function

$$C_\beta(y-x) := \sum_{N=0}^{\infty} \beta^N \, |\mathcal{S}_N(y,x)|,$$

defined for all values of β for which the power series is absolutely convergent. By translation invariance, $C_\beta(y-x)$ depends only on the displacement vector $y - x$.

Definition 5.6 (SAW Susceptibility)**.** The **susceptibility** $\chi(\beta)$ of self-avoiding walks is the sum of the two-point correlation function over all end-points:

$$\chi(\beta) := \sum_{x \in \mathbb{Z}^d} C_\beta(x) = \sum_{N=0}^{\infty} c_N \, \beta^N.$$

The series converges uniformly absolutely if and only if $0 \leq |\beta| 1/\mu_d$, where $1/\mu_d$ is the critical radius of convergence.

Proposition 5.7 (Exponential Decay of the Correlation Function)**.** *Let $\beta \geq 0$ be such that $\beta < 1/\mu_d$. Then for every $\varepsilon > 0$ there exists a constant $K_\varepsilon < \infty$, independent of x, such that*

$$C_\beta(x) \;\leq\; K_\varepsilon \big((\mu_d + \varepsilon)\beta\big)^{|x|} \qquad \textit{for all } x \in \mathbb{Z}^d.$$

Proof. We know that $|\mathcal{S}_N(0,x)| = 0$ whenever $N < |x|$, and the sum defining $C_\beta(x)$ may be restricted to $N \geq |x|$:

$$C_\beta(x) = \sum_{N=|x|}^{\infty} \beta^N \, |\mathcal{S}_N(x,0)| \;\leq\; \sum_{N=|x|}^{\infty} \beta^N c_N. \tag{5.1}$$

Fix $\varepsilon > 0$. Since $\mu_d = \lim_{N\to\infty} c_N^{1/N}$, by the definition of limit there exists $N_0 \in \mathbb{N}$ such that

$$c_N^{1/N} < \mu_d + \varepsilon \qquad \text{for all } N \geq N_0,$$

that is,

$$c_N < (\mu_d + \varepsilon)^N \qquad \text{for all } N \geq N_0.$$

For $N = 0, 1, \ldots, N_0 - 1$, define

$$K_\varepsilon \;:=\; \max\left(1,\ \max_{0\leq N<N_0} \frac{c_N}{(\mu_d+\varepsilon)^N}\right),$$

which is finite, being the maximum of finitely many finite quantities. By construction it holds:

$$c_N \;\leq\; K_\varepsilon(\mu_d+\varepsilon)^N \qquad \text{for all } N \geq 0.$$

Combining it with (4.1) we obtain

$$C_\beta(x) \;\leq\; K_\varepsilon \sum_{N=|x|}^{\infty} \big[(\mu_d+\varepsilon)\beta\big]^N.$$

Choosing ε small enough that $r := (\mu_d+\varepsilon)\beta < 1$ (possible since $\beta < 1/\mu_d$), the tail of the geometric series equals

$$\sum_{N=|x|}^{\infty} r^N = \frac{r^{|x|}}{1-r}.$$

Hence

$$C_\beta(x) \;\leq\; \frac{K_\varepsilon}{1-(\mu_d+\varepsilon)\beta}\big[(\mu_d+\varepsilon)\beta\big]^{|x|}.$$

where $\dfrac{K_\varepsilon}{1-(\mu_d+\varepsilon)\beta}$ is still finite and x-independent. $\square$

5.2 SAW representation of G

The proof of the theorem 3.1 is similar to the one given by Schenker in []. We are going to work with the Green's function defined on a volume where one site is removed every stage. Therefore , for any $\Lambda \supset \mathbb{Z}^d$ (finite or infinite) the Green function changes:

$$G^\Lambda(x,y;z) := \begin{cases} (H_\omega^\Lambda - z), & \forall x,y\in\Lambda \\ 0 & \text{, otherwise} \end{cases} \tag{5.2}$$

where $H_\omega^\Lambda - z \in \mathbb{C}^{\Lambda\times\Lambda}$is the matrix obtained by restricting $H_\omega - z$ to Λ. In this thesis we will consider $\Lambda = \mathbb{Z}^d$, but we leave the noation Λ to highligth the fact that the same result holds for any volume.
To notice is also that $H_\omega^\Lambda - z$ is still invertible for all $z \in \mathbb{C}\setminus\mathbb{R}$. In this section, for semplicity during calculations, we will write $G^\Lambda(x,y)$ instead of $G^\Lambda(x,y;z)$.

Theorem 5.8. *Let $x,y\in\mathbb{Z}^d$, $x\neq y$, $\Lambda\subseteq\mathbb{Z}^d$. G^Λ is defined as in (4.1). It holds*

$$G^\Lambda(x,y) = \sum_{n=1}^{N}\sum_{\gamma\in\mathcal{S}_n(x,y)}\prod_{i=1}^{n} G^\Lambda(x,x)\cdot G^{\Lambda_i}(\gamma_i,\gamma_i) + R_N \tag{5.3}$$

where $\Lambda_i = \Lambda\setminus\{x,\gamma_1,\dots,\gamma_{i-1}\}$ and

$$R_N = \sum_{x\sim y}\sum_{\gamma\in\mathcal{S}_N(0,x)}\prod_{i=1}^{N-1} G^\Lambda(x,x)\cdot G^{\Lambda_i}(\gamma_i,\gamma_i)\cdot G^{\Lambda_N}(\gamma_N,y). \tag{5.4}$$

Claim 5.9.

$$G^\Lambda(x,y) = G^\Lambda(x,x)\sum_{x_1\sim x} G^{\Lambda_1}(x_1,y) \tag{5.5}$$

Proof. Using the resolvent identity for $A = (H_\omega^\Lambda - z)$ and $B = (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda\setminus\{x\}} - z)$, we obtain:

$$\begin{aligned}&(H_\omega^\Lambda - z)^{-1} - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda\setminus\{x\}} - z)^{-1} \\ &= -(H_\omega^\Lambda - z)^{-1}\left(H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda\setminus\{x\}})\right)(H_\omega^{\{x\}} \oplus H_\omega^{\Lambda\setminus\{x\}} - z)^{-1}\end{aligned}$$

We know:

$$\begin{aligned} H_\omega^\Lambda &= 2d - \sum_{x_1\sim y}\delta_{x_1} + \lambda V_\omega \quad \forall y\in\Lambda \\ H_\omega^{\{x\}} &= \lambda\omega_x + 2d \\ H_\omega^{\Lambda\setminus\{x\}} &= 2d - \sum_{x_1\sim y}\delta_{x_1} + \lambda V_\omega - \lambda\omega_x \quad \forall y\in\Lambda\setminus\{x\}\end{aligned}$$

$$\implies H^\Lambda_\omega - (H^{\{x\}}_\omega \oplus H^{\Lambda\setminus\{x\}}_\omega) = -\sum_{x_1\sim x} \delta_{x_1} =: T_{x,x_1}$$

Now considering the matrix elements in (x,y) with $x \neq y$ and matrix multiplication:

$$(H^\Lambda_\omega - z)^{-1}_{x,y} - 0 = -\sum_{x_1,x_2} (H^\Lambda_\omega - z)^{-1}_{x,x_1} T_{x_1,x_2} (H^{\{x\}}_\omega \oplus H^{\Lambda\setminus\{x\}}_\omega - z)^{-1}_{x_2,y}$$

For $T_{x_1,x_2} \neq 0$, we need $x_1 = x$ and $x_2 \sim x_1$.

We obtain:

$$G^\Lambda(x,y) = \sum_{x_1\sim x} G^\Lambda(x,x) G^{\Lambda\setminus\{x\}}(x_1,y)$$

□

Theorem 5.8. Now we separate the case in which $x_1 = y$ (which implies $|x-y| = 1$).

$$G^\Lambda(x,y) = G^\Lambda(x,x) \left(\sum_{x_1\sim x, x_1\neq y} G^{\Lambda_1}(x_1,y) + I_{|x-y|=1}\, G^{\Lambda_1}(y,y) \right)$$

Iterating N times, it gives (5.3). □

5.3 Proof with SAW

Now we use self-avoiding-walks to prove the theorem (3.1).

Proof. We take the average of (4.2), and using $|\sum a|^s \leq \sum |a|^s$ (which follows from triangle inequality), we have:

$$\mathbb{E}\left[|G^\Lambda(x,y)|^s\right] \leq \sum_{n=1}^{N} \sum_{\gamma\in\mathcal{S}_n(x,y)} \mathbb{E}\left[|G^\Lambda(x,x)|^s \prod_{i=1}^{n} |G^{\Lambda_i}(\gamma_i,\gamma_i)|^s \right] + \tag{5.6}$$

$$\sum_{x\sim y} \sum_{\gamma\in\mathcal{S}_N(0,x)} \mathbb{E}[|G^\Lambda(x,x)|^s \prod_{i=1}^{N-1} |G^{\Lambda_i}(\gamma_i,\gamma_i)|^s \cdot |G^{\Lambda_N}(\gamma_N,y)|^s.] \tag{5.7}$$

The resolvent $G^{\Lambda_j}(w,w')$ does not depend on the random variables ω_{x_i}, $i = 0,\ldots,j-1$. Hence, recursive applications of the a priori bound (3.2) yield

$$\mathbb{E}\left[|G^{\Lambda}(x,x)|^s \prod_{j=1}^{n} |G^{\Lambda_j}(\gamma_j,\gamma_j)|^s\right] \leq \left(\frac{1}{1-s}\frac{1}{\lambda^s}\right)^{n+1}$$

and

$$\mathbb{E}\left[|G^{\Lambda}(x,x)|^s \prod_{j=1}^{N-1} |G^{\Lambda_j}(\gamma_j,\gamma_j)|^s \cdot |G^{\Lambda_{N-1}}(\gamma_N,y)|^s\right]$$
$$\leq \left(\frac{1}{1-s}\cdot\frac{1}{\lambda^s}\right)^N \cdot \mathbb{E}\left[|G^{\Lambda_{N-1}}(\gamma_N,y)|^s\right] \leq \left(\frac{1}{1-s}\frac{1}{\lambda^s}\right)^N \frac{1}{|\operatorname{Im}(z)|^s}$$

Where we used a priori bound and lemma 3.5.

Inserting these estimates in the sums above, we get:

$$\mathbb{E}\left(|G^{(\Lambda)}(x,y)|^s\right) \leq \sum_{n=0}^{N} \beta(s)^{1+n} |\mathcal{S}_n^{(\Lambda)}(y,x)| + Err(N)\frac{1}{|\operatorname{Im} z|^s} \tag{5.8}$$

with $Err(N) := \beta(s)^N |\mathcal{S}_N^{(\Lambda)}(x)|$ $\beta(s) = \frac{1}{1-s}\frac{1}{\lambda^s}$ Let's consider the susceptability $\chi(\beta)$ of the self avoiding walks in $\mathcal{S}_N^{(\Lambda)}(x)$:

$$\chi(\beta) = \sum_{x\in\mathbb{Z}^d}\sum_{N=0}^{\infty} \beta(s)^N\, |S_N^{(\Lambda)}(x)| = \sum_{N=0}^{\infty} c_N\, \beta(s)^N. \tag{5.9}$$

The series converges absolutely if and only if $0 \leq |\beta| < 1/\mu_d$, where $1/\mu_d = \lim_{N\to\infty} c_N^{1/N}$ is the critical radius of convergence. Thus, for $|\beta(s)| = \frac{1}{|(1-s)\lambda^s|} \leq \frac{1}{\mu_d}$, $\chi(\beta)$ converges uniformly, and most importantly:

$$Err(N) \longrightarrow 0, \quad \text{for} \quad N\to\infty$$

We send N to the infinity and obtain from (4.7):

$$\mathbb{E}\left(|G^{(\Lambda)}(x,y)|^s\right) \leq \sum_{n=0}^{\infty} \beta(s)^{1+n} |\mathcal{S}_n^{(\Lambda)}(y,x)| \tag{5.10}$$

Taking the supremum over volumes Λ now yields

$$\sup_{\Lambda\subset\mathbb{Z}^d} \mathbb{E}\left(|G^{(\Lambda)}(x,y)|^s\right) \leq \beta(s) C_{\beta(s)}(x-y). \tag{4.3}$$

It remains to establish for which λ we may find $s \in (0,1)$ with $\beta(s) < 1/\mu_d$. As above, for $\lambda > e$ the unique critical point is $s_{\text{crit}} = 1 - 1/\ln\lambda$ and

$$\min_{s\in(0,1)} \beta(s) = \beta(s_{\text{crit}}) = \frac{e\ln\lambda}{\lambda}$$

which is less than $1/\mu_d$ if and only if $\lambda > \lambda_{\text{And}}$. Finally we get:

$$\mathbb{E}\left(|G^{(\Lambda)}(x,y;z)|^{1-\frac{1}{\ln\lambda}}\right) \leq \beta(\lambda) K_\epsilon \ln\lambda\, e^{-m_\epsilon(\lambda)|x-y|} \tag{5.11}$$

where

$$m_\epsilon(\lambda) = -\ln\beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

$\square$

6 Connection to Long-Range SAW

The aim of this section is to use the strategies of the previous chapters in order to obtain similar results for long range jumps. In particular, we are comparing the two models and again applying both fractional moments method and the self-avoiding-walk representation of the Green's function. First of all we define the fractional descrete Laplacian and it's polynomial decay, then we'll compare the SAW representation with the one for the one-jump operator and finally, there will be an in-depth look at the localisation conditions in the fractional case. This chapter is based on the work of M. Disertori, R.M. Escobar and C. Rojas-Molina in 2024 [].

6.1 The fractional Laplacian

For the long-range, the kinetic operator doesn't connect only the first neighbors but also distant sites in the lattice. For this reason we need the fractional of the standard descrete Laplacian.

Definition 6.1. Let $-\Delta$ be defined as in 1.4 in Section 1. The fractional descrete Laplacian is $(-\Delta)^\alpha$ for $0 < \alpha < 1$

Just as the standard descrete Laplacian, $(-\Delta)^\alpha$ is bounded, translation invariant and satisfies

$$(-\Delta)^\alpha > 0 \quad \text{on the diagonal,} \qquad (-\Delta)^\alpha \leq 0 \quad \text{otherwise}$$

. An important result in this study is the polinomial decay of the fractional moments of the Green's function, in contrast with the exponential decay that we proved in the previous chapters. This follows, in particular, from the polinomial decay of the off-diagonal elements of the fractional Laplacian.

In fact, there exist constants $0 < c_{\alpha,d} < C_{\alpha,d}$ such that

$$\frac{c_{\alpha,d}}{|x-y|^{d+2\alpha}} \leq -(-\Delta)^\alpha_{x,y} \leq \frac{\mathcal{C}_{\alpha,d}}{|x-y|^{d+2\alpha}} \tag{6.1}$$

for all $x, y \in \mathbb{Z}^d$ and $x \neq y$.
From this bound, we obtain

$$\sum_{x\in\mathbb{Z}^d} |(-\Delta)^\alpha_{0,x}|^s < \infty, \qquad \forall\, 0 < \alpha \leq 1 \quad \text{and} \quad s \in \left(\frac{d}{d+2\alpha}, 1\right] \tag{6.2}$$

.

More precisely:

$$\sum_{|x|\geq 1,\, x\in\mathbb{Z}^d} |(-\Delta)^{\alpha}_{0,x}|^s \leq C^s_{\alpha,d} \sum_{|x|\geq 1,\, x\in\mathbb{Z}^d} \frac{1}{|x|^{(d+2\alpha)s}}.$$

By standard comparison of the lattice sum with an integral on $\mathbb{R}^d$:

$$\sum_{|x|\geq 1,\, x\in\mathbb{Z}^d} \frac{1}{|x|^p} \leq C_d \int_{|z|>1} \frac{1}{|z|^p}\, dz.$$

Passing to spherical coordinates in $\mathbb{R}^d$, the integral reads

$$\begin{aligned} \int_{|z|>1} \frac{1}{|z|^p}\, dz &= |\mathbb{S}^{d-1}| \int_1^\infty \frac{r^{d-1}}{r^p}\, dr \\ &= |\mathbb{S}^{d-1}| \int_1^\infty r^{d-1-p}\, dr, \end{aligned}$$

where $|\mathbb{S}^{d-1}| = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ denotes the surface area of the unit sphere in $\mathbb{R}^d$.

The radial integral converges if and only if the power satisfies

$$d-1-p < -1 \iff p > d.$$

Substituting $p = s(d+2\alpha)$, this condition becomes

$$s(d+2\alpha) > d \iff s > \frac{d}{d+2\alpha}.$$

Since $\alpha > 0$ implies $d+2\alpha > d$; for every $s \in \left(\frac{d}{d+2\alpha}, 1\right]$, we conclude that

$$\sum_{x\neq 0} |(-\Delta)^{\alpha}_{0,x}|^s \leq C^s_{\alpha,d} \sum_{x\neq 0} \frac{1}{|x|^{s(d+2\alpha)}} < \infty.$$

6.2 The SAW representation

As well as in the standard case, we consider the Hamiltonian operator with his random potential term λV_ω and the fractional kinetic term $(-\Delta)^\alpha$. This results consider an arbitrary continuous distribution with compact supported density.

The Green's function is again the resolvent and its decay is close to the decay of the fractional Laplacian. In fact, the averaged fractional Green's function satisfies

$$\mathbb{E}[|G_z(x,y)|^s] \leq \frac{K_0\theta_s}{\lambda^s}\frac{1}{|x-y|^{d+2\alpha_s}} \qquad \forall\, x,y\in\mathbb{Z}^d,\ x\neq y, \tag{6.3}$$

uniformly in $z \in \mathbb{C}\setminus\mathbb{R}$, for all $\lambda > \lambda_0(s) := (\frac{\theta_s}{R})^{\frac{1}{s}}$, where $R_{\chi(\beta,\alpha)}$ is the radius of convergence of the susceptability and with $\alpha_s := s(d+2\alpha)-d$. In addition, $K_0 = K_0(d,\alpha)$ comes from the two-point function bound and $\frac{\theta_s}{\lambda^s}$ comes from the *a priori bound.*

Remark 6.2. Notice that for $\alpha = 1$ and considering the uniform distribution in $[-1,1]$ for ω, $\theta_s = \frac{1}{1-s}$ which leads to $\lambda_0(s) = \lambda_{\text{And}}$.

In the long-range scenario, we are still using the same proof techniques as in Chapter 5 in order to prove (6.3).
Nonetheless, one important change is given by the fact that, for long jumps, we are not just considering the neighbors, but an arbitrary site on the lattice. In particular, the probability to jump from the site x to the site y changes into:

$$p(x,y) := \frac{|(-\Delta)^\alpha_{x,y}|^s}{\sum_{z\neq x}|(-\Delta)^\alpha_{x,z}|^s}.$$

Using the random walk on $\mathbb{Z}^d$ with $p(x,y)$ as transition probability, changes the definitions given in Chapter 5.1, which are now all generated by $|(-\Delta)^\alpha_{x,y}|^s$ and well defined for the values of s as in (6.2).

In order to prove (6.3) we use the same proof steps as before:

1. As in Claim 5.9, by the resolvent identity we obtain:

$$G_z^\Lambda(x,y) = G_z^\Lambda(x,x)\sum_{x_1\in\Lambda\setminus\{x\}}\Delta^\alpha_{x,x_1}G^{\Lambda_1}(x_1,y) \tag{6.4}$$

   with $\Delta^\alpha = -(-\Delta)^\alpha$ greater than zero for the off-diagonal elements. Since we are working with $0<\alpha<1$, we don't obtain an operator such as T_{x,x_1} but we keep the Laplacian fractional.

2. We then iterate and bound like in Chapter 5 for (5.8), in order to get:

$$\mathbb{E}[|G_z^\Lambda(x,y)|^s] \leq \frac{\theta_s}{\lambda^s}C_\beta^{\alpha,s}(y-x) + \frac{1}{|\operatorname{Im} z|^s}\mathit{Err}(N). \tag{6.5}$$

   In this case, $\mathit{Err}(N)$ depends on $|(-\Delta)^\alpha_{x,y}|^s$ and under the conditions of (6.2) it vanishes for $N \to \infty$, solving the problem of the $\operatorname{Im} z$ as denominator in the equation.

3. Finally, (see [Lemma 4 of PAPER DISERTORI]),

$$C_{\beta}^{\alpha,s}(y-x) \leq K_0 \frac{1}{|y-x|^{d+2\alpha}} \tag{6.6}$$

holds as a consequence of (6.1).

6.3 The Localisation critiria

A natural question that may arise from the previuous statements is how localisation is proven without exponental decay of the fractional moments of the Green's function. In the long-range case it is only possible to prove a weaker version of dynamical localisation. Without finitness for all moments of the position operator, it is convenient to prove spectral localisation via Simon-Wolff theorem.

Lemma 6.3. *Let's* (6.3) *hold uniformly in* $z \in \mathbb{C} \setminus \mathbb{R}$ *and for* $\lambda > \lambda_0(s)$. *Then it follows:*

(i) The spectrum of $H_{\alpha,\omega}$ *consists only of pure point spectrum for a.e.* $\omega \in \Omega$

(ii) for $0 < \beta < 2\alpha$ *we have for a.e.* $\omega \in \Omega$

$$\sup_{t \in \mathbb{R}} \left\| |x|^{\frac{\beta}{2}} e^{-itH_{\alpha,\omega}}(0,x) \right\| < \infty. \tag{6.7}$$

In particular, if $\alpha > \frac{1}{2}$ *the first moment of the wave function initially localized at the origin and evolving under the action of* $H_{\alpha,\omega}$ *is finite.*

In fact, for $0 < \beta < 2\alpha$ we have for a.e. $\omega \in \Omega$

$$\sup_{t \in \mathbb{R}} \left\| |x|^{\frac{\beta}{2}} e^{-itH_{\alpha,\omega}}(0,x) \right\| < \infty. \tag{6.8}$$

In particular, if $\alpha > \frac{1}{2}$ the first moment of the wave function initially localized at the origin and evolving under the action of $H_{\alpha,\omega}$ is finite.